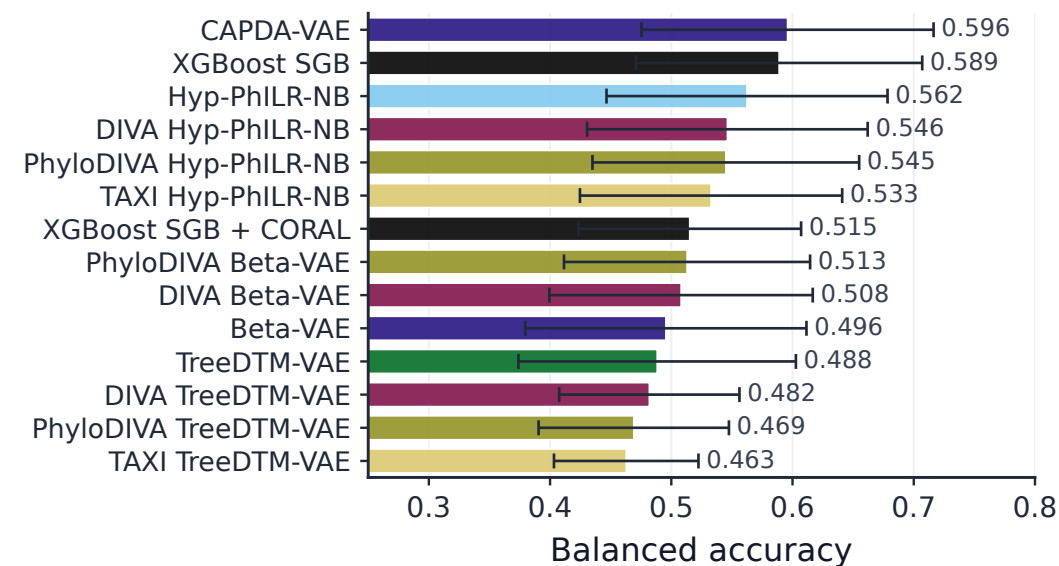


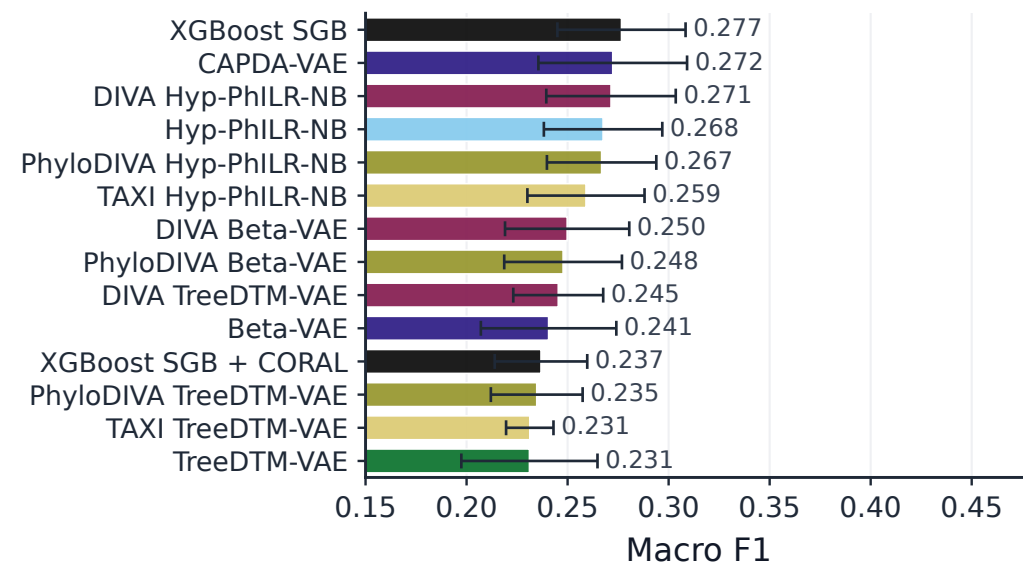
A

Strict LOSO balanced accuracy



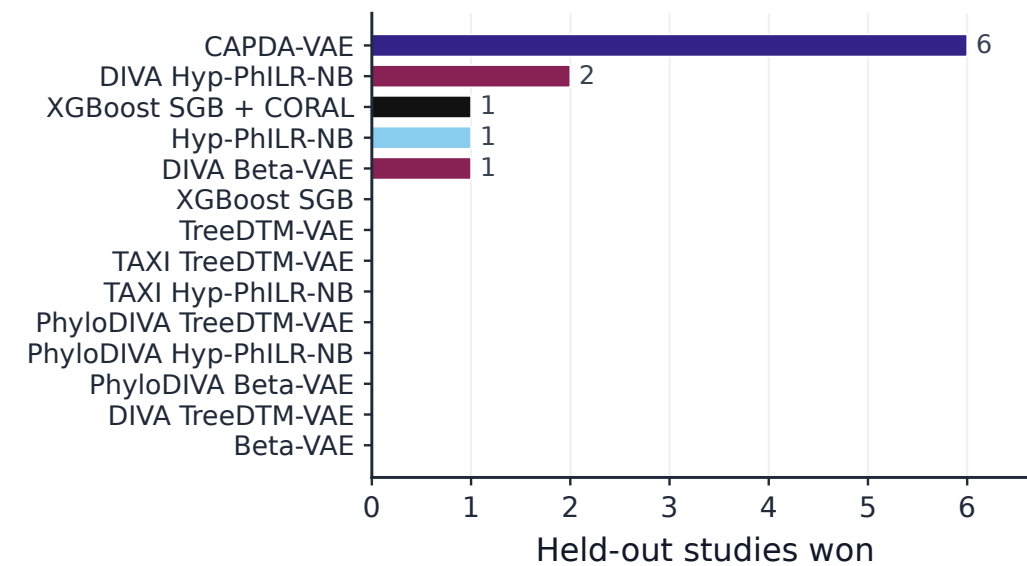
B

Strict LOSO macro F1



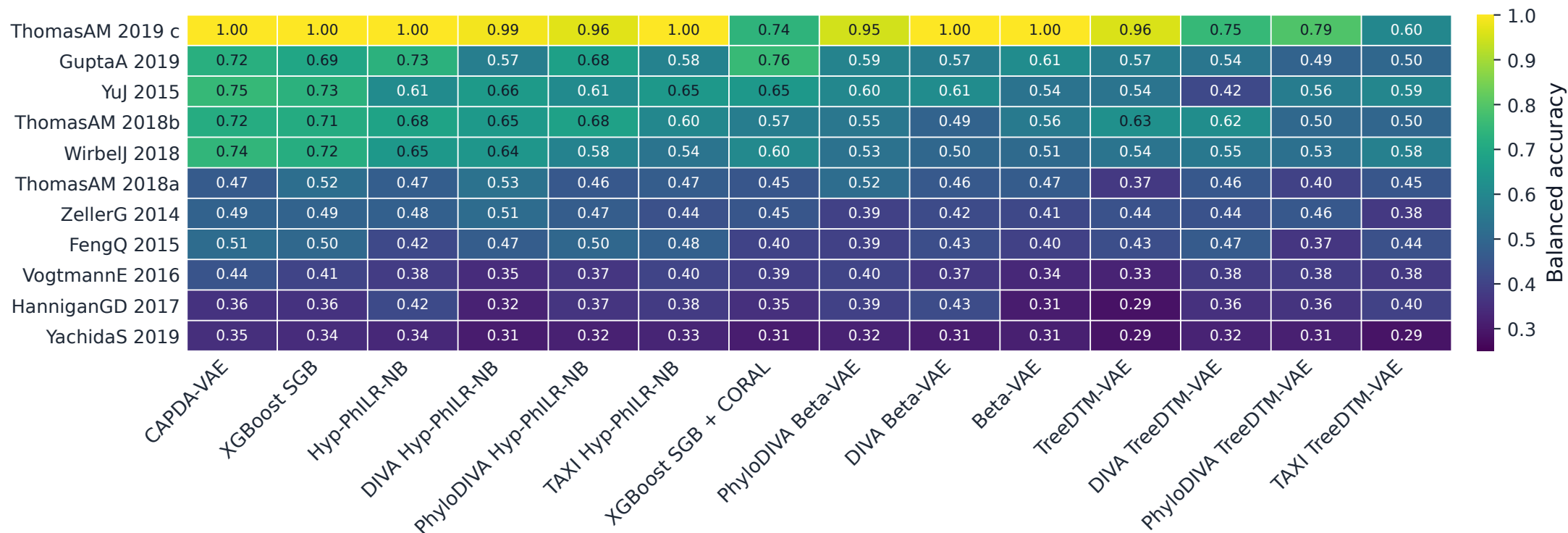
C

Best-model frequency



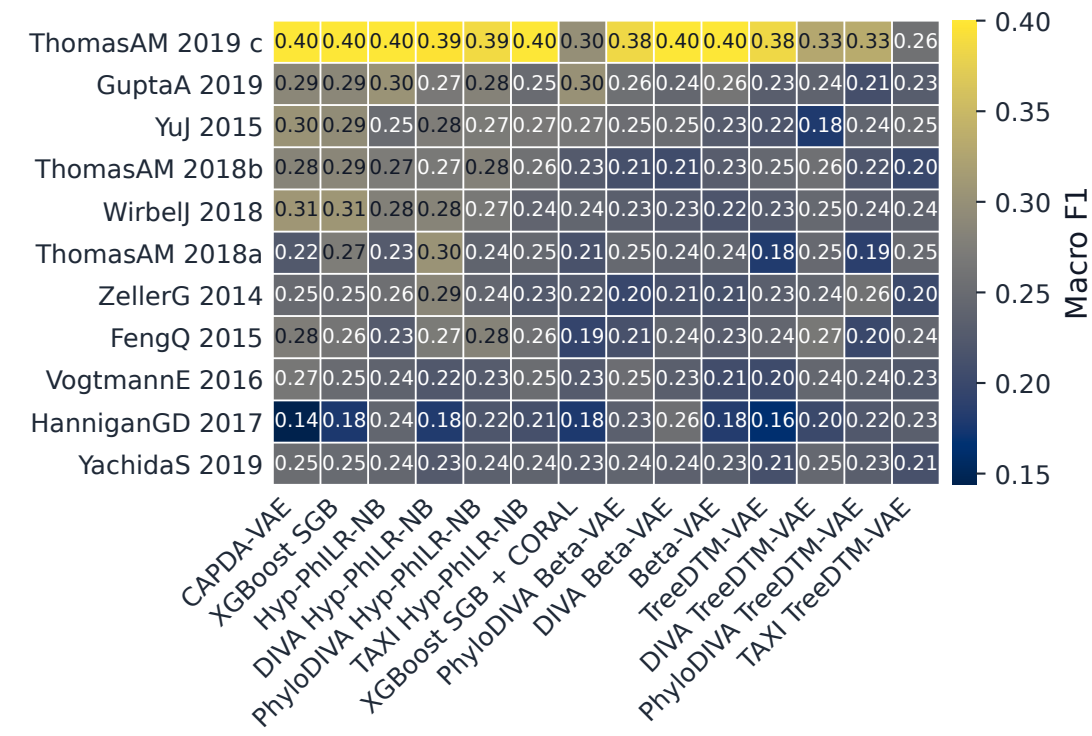
D

Balanced accuracy by held-out study



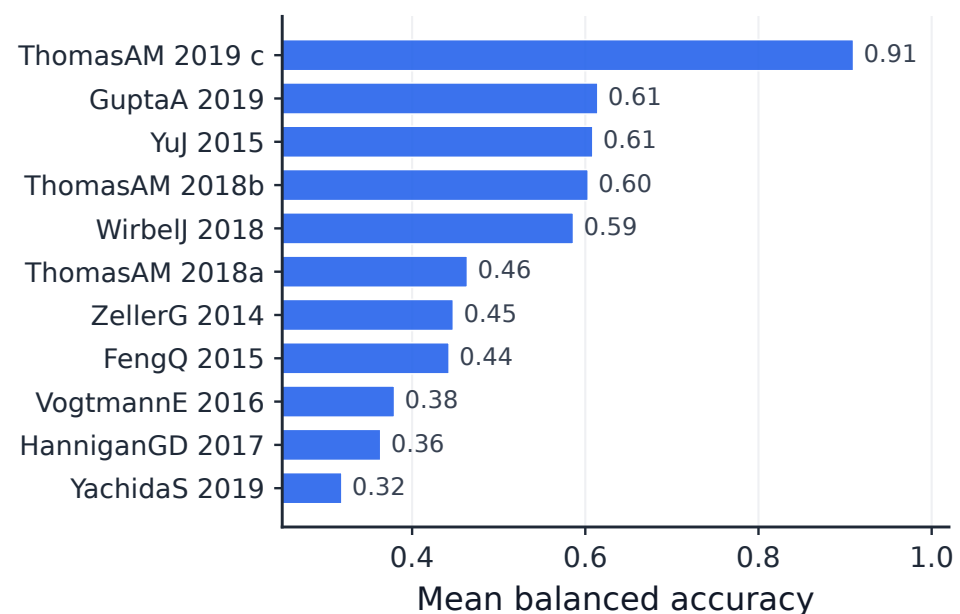
E

Macro F1 by held-out study



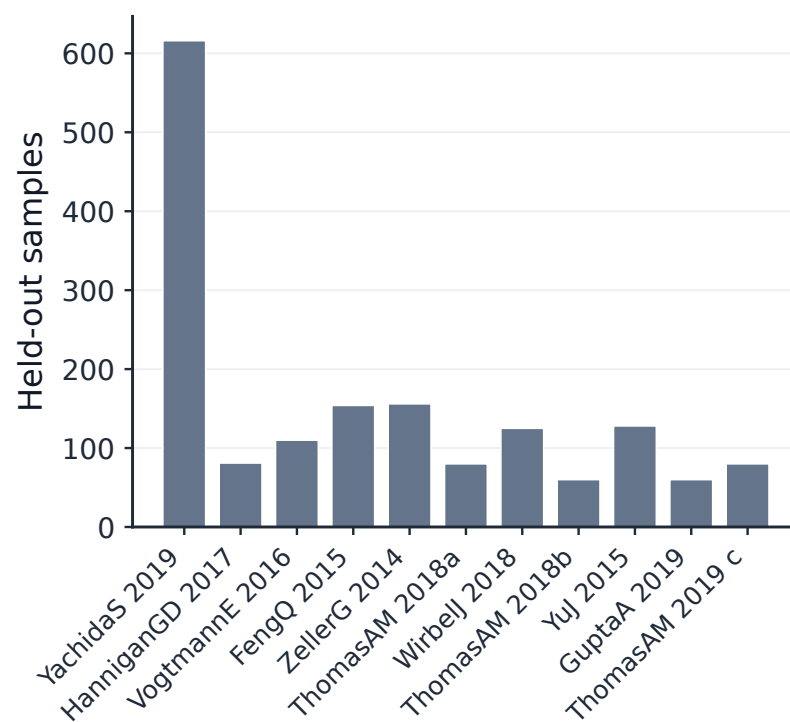
F

Held-out cohort difficulty



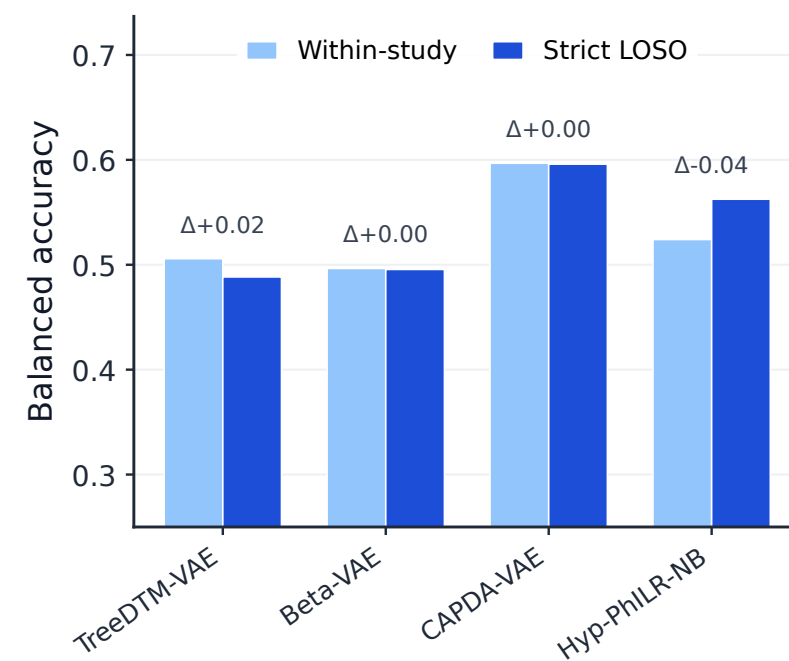
G

Evaluation cohort size



H

Within-study → LOSO transfer gap



Model family

- Vanilla VAE
- PhILR
- PhyloFusion
- Taxonomy-aware
- Tree-structured
- Baseline (XGBoost)
- DIVA